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Interpreting for the agri-food sector: A gamified approach

Abstract: The agri-food sector plays an important role in the economic system of the European Union (and particularly Spain). In this regard, international exports of agri-food products are becoming increasingly significant, making document translation and interpreting necessary in this area. However, little attention has been paid to this branch of specialization in Translation and Interpreting University curricula. In this chapter we present a protocolized gamified methodology to prepare for an interpretation on this particular domain. We will focus on the acquisition of domain-specific terminology through gamification, corpus-driven and technology-mediated stance. After providing a brief overview on the agri-food sector and its multilingual communication needs, we will focus on interpreter preparation. Our methodology draws on gamification for a better learning experience. It is a three-step corpus-based protocol for the creation of terminology-orientated game-based activities. To the best of our knowledge, this is one of the first studies on the use of gamification to prepare for an interpretation (real or simulated).

Keywords: Gamification, agri-food sector, professional interpreter, trainee interpreter, terminology preparation

1 Introduction

In an environment of climate crisis characterized by escalating pollution and the problem of climate change, the flourishing agri-food industry appears to be on the spotlight worldwide. Being a fundamental part of the economy of all countries, this sector faces several challenges nowadays, such as the need to improve the sustain-

Acknowledgements: This paper has been carried out in the framework of various research projects on language technologies applied to translation and interpretation (ref. PID2020-112818GB-I00, PDC2021-121220-I00, TED2021-129789B-I00, and ProyExcel_00540). It has also been funded by a FPU (Formación del Profesorado Universitario) contract, ref. no. FPU21/03104, awarded by the Spanish Ministry of Science, Innovation and Universities to Marta Alcaide-Martínez.

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ability and quality of food while reducing its consumption of natural resources and the use of inefficient and costly processes. This lies at the root of current attempts to apply artificial intelligence (AI) in this sector, as several AI techniques (e.g., machine learning, deep learning or artificial neural networks) could help automate tasks, save time, improve sustainability and product quality, and could even contribute to lighten linguistic mediation tasks, like translation and interpreting, in a more cost-effective way for companies (Rejeb et al., 2022).

In the case of Spain, the agri-food sector has been traditionally considered a strategic production activity (Fernández Núñez, 2000), mainly due to the steady rise of exports, especially to the European Union (Medina Reguera & Álvarez García, 2014). As Álvarez García (2018) highlights, based on statistics from the Spanish Instituto de Comercio Exterior, exports are having an increasing impact on the Spanish economy, and this also has a direct repercussion on an upsurge in the demand for language-mediated tasks.

The same rising trend can be observed with regards to translation and interpreting demands for commercial purposes. In these commercial exchanges, it is common to use the language of the exporting or importing country or a third agreed language as *lingua franca*. A clear example is the wine sector in Spain. The study by Ibáñez Rodríguez et al. (2010) reveals a tendency to use English as *lingua franca* for international communication, although most respondents consider translation to be very or fairly important for the performance of their commercial activity in foreign countries. Interestingly enough, the wine industry also tends to dispense with professional translators and interpreters and rely instead on in-house staff, as technical expertise in the sector is considered more important than linguistic competence (Ibáñez Rodríguez et al., 2010; Medina Reguera & Álvarez García, 2014).

In this chapter we intend to make a significant contribution to the didactics of Translation and Interpreting, with special reference to interpreters. Against this background, our research focuses on the potential of gamification to assist interpreters working in the agri-food sector in their advance preparation phase of a given job assignment. With this aim, we present a variety of gamified activities that will allow professional interpreters and trainees to practice their mastery of domain-specific terminology in an innovative, entertaining, and meaning way.

The rest of the chapter is organized as follows. Section 2 is composed of two subsections dealing with gamification, some key concepts and its benefits, as well as the advantages of applying it to translation and, especially, to the preparation phase of an interpretation. In Section 3, we explain the methodology, the target audience, the compilation of our corpus and the extraction stage of seed terms for the gamified activities that we have developed on two open access gamification platforms and that we have described in Section 4. In Section 5 we state our

conclusions and in Section 6 we cite all the bibliographical sources that underpin our chapter. To the best of our knowledge, this is one of the first studies on the use of gamification to prepare for an interpretation (real or simulated).

2 The advantages of game-based interpreter preparation

Domain-specific terminology preparation is considered central to improving the quality of interpretation, as already argued by Díaz-Galaz (2011), Pöchlacker (2016), Fantinuoli (2017) and Xu (2018), among others. Interpreters usually acquire domain and terminology knowledge in the preparation phase, as an anticipatory strategy to support the interpreting event and overcome the inherent difficulties of the interpreting process (cf. Pöchlacker, 2016). Typical preparatory activities include glossary compilation, bilingual lists, and terminology management (Díaz-Galaz, 2011; Chang, Wu, & Kuo, 2018). However, these techniques have a number of disadvantages that could be avoided by using gamification as a preparation method for interpreters, as it will be explained in the following subsections.

2.1 Gamification: Key concepts and benefits

The term *gamification* has its origins in the digital media sector, although it has been spreading and becoming popular in other contexts, such as healthcare, labor, education, tourism, etc. This term was coined by Nick Pelling, a British videogame programmer, in 2002, but its first documented uses were not until 2008 in a blog post by Brett Terril. Since then, the concept of gamification has been growing to achieve the scope and importance it has in today's society due to the multiple advantages of its implementation in the various fields in which it has been investigated (Deterding et al., 2011, Contreras Espinosa & Eguia, 2017). It is worth noting that gamification, under another denomination, has been applied and studied by other authors before, such as Ellenor Fenn in the eighteenth century, whom Gallego-Sturla and Taillefer (2021) consider a precursor of ludology, currently better known as gamification. Despite this, it was not until mid-2010 that the notion began to gain more relevance and to be defined by the academic world, with the definition by Deterding et al. being one of the most widely accepted and cited: “Gamification is the use of game design elements in non-game contexts” (2011, p. 2).

From this definition, we extract a series of key concepts that are essential to distinguish and understand in depth when creating gamified activities. The concept

of *game* is different from *gamification*, as *games* focus on making users entertained and having fun (Xu et al., 2017). Gamification involves meaningful learning and real problem solving, while in games fictional situations are more usual. Moreover, in their paper, Alcaide-Martínez and Taillefer (2022) expose the definitions of the terms *play* and *serious games*. They emphasize that *play* seeks the amusement of the participant without the need to use rules, since enjoyment is the important thing. As with games, in plays the user does not face real problems. *Serious games*, on the other hand, focus on individual experience, take place in a specific space and context and are related to the acquisition of knowledge and skills (Kapp, 2012; Xu et al., 2017). Therefore, gamification is not a game as such, but a set of fused elements that aim to evoke the feelings provoked and the skills awakened by games, plays and serious games.

The *elements*, mentioned in the definition of Deterding et al. (2011), are also crucial in the design of gamified scenarios and activities. Werbach and Hunter (2015) divide these elements into three categories: dynamics, mechanics, and components. Hence, it is necessary to take into consideration that the dynamics constitute the general elements that are not directly introduced in the gamified system. The mechanics are the processes and procedures that involve the player in the activity and encourage engagement between the user and the gamified exercise. Finally, the components are the specific parts and accessories of the games that, correctly combined with the mechanics and dynamics, constitute the gamified structure (Werbach & Hunter, 2015). In the conception of the gamified activity, the appropriate elements should be chosen according to the target audience of the given exercise and the objectives to be achieved, as detailed in the description of our activities designed for professional interpreters and trainees (cf. the pyramidal classification of dynamics, mechanics, and components in Alcaide-Martínez and Taillefer, 2022, p. 286).

Cornellà et al. (2020) state in their paper that game is a cornerstone in the development and learning of human beings. Unfortunately, it is not usually used in the educational world because it is associated with leisure, so it is often reserved for early childhood education. However, these authors argue that game has great educational potential, since its entertaining, relaxed and fun nature fosters the acquisition of significant theoretical and practical learning. A properly designed or selected gamified exercise enables students to increase their attention and effort, memorize and learn in a rewarding and amusing way, as opposed to traditional and more serious methodologies. Likewise, gamification should not be confused with Game Based Learning (GBL).

As Cornellà et al. (2020) point out, gamification consists of the creation of a scenario characterized by game elements in which users/students are the protagonists and have to achieve a proposed challenge, while GBL is the use of games so

that users/students learn something through them. Both are active methodologies that either separately or in combination would have many advantages for the learning process and the process of preparing for a possible interpreting assignment, because games and their elements generate stimulation, promote concentration, awaken curiosity, elicit feelings, enable teamwork and relationships with peers, increase motivation and the ability to react in order to solve problems, decrease stress, teach how to use information effectively, make it possible for participants to face real problems and situations, etc. (Marín et al., 2015). All these positive aspects could be very beneficial not only for interpreting students, but also for professional interpreters, especially in the preparation phase, which is so important for the quality of an interpreting (Díaz-Galaz, 2011; Pöchhacker, 2016; Fantinuoli, 2017; and Xu, 2018). Gamification would help, for example, to lighten the tedious task of collecting and memorizing domain-specific terminology.

2.2 Gamification in translation and interpreting

Among the scientific productions on gamification, we have no evidence that a three-way connection between gamification, interpreting and agri-food has been previously drawn. Nonetheless, we have located studies linking gamification with translation and interpreting, such as those by Gutiérrez-Artacho and Olvera-Lobo (2016), Alcalde Peñalver and Santamaría Urbieto (2020), Alcaide-Martínez and Taillefer (2022), among others. Despite these works, gamification has not been frequently applied in the aforementioned disciplines or in the university degree. In this case, our focus will be on interpreting, as it is well known that translation students and professional translators have many more technological tools at their disposal for their preparation and professional development.

As mentioned at the beginning of this section, advance preparation is crucial in all types of interpreting (simultaneous interpreting, conference interpreting, etc.). Díaz-Galaz et al. (2015) remark that prior documentation and preparation is an indispensable activity for interpreting students and professionals, as it allows them to anticipate the speech and, thus, experience lower levels of tension and stress. Both students and professionals dedicate plenty of time to this preparation phase and, normally, the result of this invested time is a glossary with specialized terminology in the two working languages. Depending on the interpreter, glossaries can simply be a list of terms in the source language and their equivalents or a bilingual list together with more complementary information that offers more insight into the different concepts.

However, glossaries and bilingual lists have several disadvantages: the interpreter has to invest a lot of time in their compilation, they are not interactive or visually attractive tools, the only way to work with this resource is memorization and repetition, they are more useful for translators than for interpreters because of their textual nature, these glossaries and lists are not eye-catching and do not arouse curiosity or interest, they are not adapted to the fast pace of modern society, etc. Apart from that, attempts to develop technological tools for interpreters include Intragloss for Mac OS, Terminus or LookUp for Windows, InterpretBank for Windows and Android, Interpreters' Help accessible via the Internet, and so on. These tools also have some limitations, such as that they cannot handle documents, they cannot include meta-information, terminology processing has to be done manually, most of them only allow the creation of bilingual glossaries, etc. (Corpas Pastor, 2018).

All the disadvantages of traditional training methods lead us to think of gamification as an alternative and innovative methodology with numerous advantages, as mentioned in the previous subsection. In addition, Corpas Pastor (2022) highlights that the prior preparation phase is especially relevant for interpreters when the subject matter is technical and specialized, since the terminology, phraseology and syntactic constructions will be more complex, as in the case of agri-food. This author also indicates that this complexity is compounded by the fact that interpreters usually have little time or no time to prepare and memorize domain-specific terminology. This is a problem that must be solved, since the mastery of domain-specific and specialized terminology and phraseology has a direct impact on the quality of the interpretation (Corpas Pastor, 2022). In fact, as stated by Álvarez García (2018) in a survey to Andalusian exporting companies, knowledge and mastery of specific, specialized and technical terminology is the aspect they regard as the most fundamental in the profile of an interpreter who is going to participate in an international negotiation.

Hassan (2017) outlines that both intrinsic and extrinsic stress is a phenomenon that interpreters often experience and that they have to learn to manage successfully. In this scenario, gamification is presented as a technological methodology that could facilitate the work of professional interpreters and trainees who have to cope with specialized agri-food discourse in a real or simulated context. One of the perks of gamified approaches is that they enable participants to suffer less stress (Patricio et al., 2022). Additionally, Fajri et al. (2021) conclude in their paper that gamification not only boosts learners' motivation, but also makes it possible to reduce technostress, which is the tension and stress produced by having to use and interact with technologies. Equally, gamification makes participants more involved and focused on the exercise, allows them to work on a specific skill in a meaningful and more fun way, increases users' confidence and

academic achievement, etc. (Dehghanzadeh & Dehghanzadeh, 2020; Manzano-León et al., 2021). Actually, Nistor and Iacob (2018) emphasize that bringing gamification into all levels of education is more of a necessity than an option.

Besides, González Cabanach (1997) already noted that memorizing superficial information can be a tedious, boring and monotonous activity for both the student of interpreting and the professional interpreter. The memorization of long lists of specialized terminology in two languages in a short period of time would be stored in the short-term memory and would not constitute meaningful learning, especially when this vocabulary is part of a technical subject, as happens with the agri-food topic. Thus, in this chapter we propose the use of gamification as an active preparation methodology for professional interpreters and trainees, since it will allow them to memorize specific simple and multi-word terminology most frequently employed in the agri-food sector in an engaging, fast and effective way by means of gamified dynamics, mechanics, and components. In this way, interpreting students and professionals will be more motivated than when they have to memorize endless bilingual lists and this will lead them to unconsciously learn domain-specific terminology and become familiar with complex syntactic structures of the subject at hand. Lastly, to exploit all these advantages it is paramount to properly design gamified activities and not reduce them only to rewards, points, and badges, since gamification involves much more than these three elements (Kapp, 2012).

3 Methodology

Despite the enormous potential of gamification, it has been rarely used to prepare for an interpretation, let alone for job assignments in the agri-food sector. For this reason, the main aim of this chapter is to develop corpus-based gamified activities on open access platforms in order to provide interpreters (professionals and trainees) a better learning experience when preparing for an interpretation within the agri-food sector. We advance the use of game-based exercises as a better alternative to traditional lists of equivalents. Our focus is terminology preparation (including multi-word terms) through gamified activities as a more enjoyable and meaningful way to acquire domain-specific knowledge for a given interpretation assignment related to the agri-food industry.

Our protocolized methodology comprises three phases: (i) compilation of a domain-specific corpus on agri-foods; (ii) extraction of gaming seeds (e.g., simple/multi-word terms and sentence concordances); and (iii) creation of seed-based exercises on two open access gamification platforms. In this section we cover the

first two phases of our methodological approach, including intended target audience, data collection and analysis and choice of corpus management system. Gamified activities designed and gamification platforms selected are presented in Section 4.

3.1 Target audience

The gamified activities created in this research are intended especially for professional and trainee interpreters. In the case of professional interpreters, these exercises could be helpful for the preparation phase of interpreters engaged in any of the existing types and modes of interpreting: conference interpreting, simultaneous interpreting, consecutive interpreting, remote interpreting, etc. However, it should be underlined that they would be particularly beneficial for simultaneous interpreting, as interpreters have less time to react and have to be very well prepared because they will most likely not have enough time to do word searches.

Regarding trainee interpreters, these exercises would be especially useful from the second or third year of the Bachelor's Degree in Translation and Interpreting, as it is when students begin to take subjects related to interpreting. As we mentioned for professional interpreters, these gamified activities could be put into practice in the modules of bilateral interpreting, conference interpreting, simultaneous interpreting or specialized interpreting. Hence, the socio-demographic characteristics of target students would be as follows: age range between 20–24 years old, native knowledge of the Spanish language and at least a B2 level of English.

3.2 Corpus compilation

For the task at hand in this chapter, we have compiled an ad hoc corpus via Sketch Engine (see below).¹ An ad hoc corpus is a virtual corpus that is compiled for a specific assignment and is usually characterized by its quality rather than its quantity of texts. This type of corpus can be created fairly quickly, and it helps translators/interpreters to meet their documentation needs in relation to an assignment on the same subject matter (Corpas Pastor, 2004). We also chose to create our own ad hoc monolingual corpus in English in order to take advantage of the benefits that Aston (2002) points out in his work: knowing the corpus in depth

¹ https://app.sketchengine.eu/#dashboard?corpname=user%2FMartaAlcaide%2Fagri_food_corpus (Retrieved March 3, 2023).

and knowing how to control it, being aware of its positive aspects and its limitations, and having the knowledge of how to exploit it appropriately. We selected texts that deal with agri-food from different perspectives and in various economic systems. Moreover, to further delimit the field of our corpus, we applied design criteria adapted from Varela Vila (2009):

- Size: we have elaborated a corpus consisting of 45 documents and a total of 601,438 tokens, 431,674 words, and 21,730 sentences. Our corpus is not excessively large, but it can be deemed fit for our purposes.
- Specialization: the compiled corpus is of high quality, since it is made up of scientific papers extracted from Google Scholar and published in journals indexed in quality databases. We opted not to modify the publications, so we have introduced in Sketch Engine the documents in PDF format, since interpreters are sometimes provided with the speaker's bibliography, related documents, etc. when they are hired for an interpretation; data that could also be useful.
- Topic: we selected scientific publications that examine the agri-food industry from different approaches and in different cultures (Egyptian, Italian, etc.). All are domain-specific and well-documented. Specialized and technical terminology from the agri-food sector can be found in all the texts.
- Date: the scientific publications that we included in the corpus are recent and range from 2005 to 2022. We did not incorporate documents prior to 2005 in order to obtain the most frequent specialized terminology currently used in this sector.
- Language variety: we have selected documents written in English, regardless of the authors' place of origin. Due to the processes of internationalization and globalization in which we are immersed, we considered it convenient to include several English varieties so that interpreters can recognize domain-specific terms, even if they are diatopically restricted.
- Genre: our corpus is composed of specialized scientific papers covering agri-food in relation to new technologies, sustainability, resource waste, supply chain, etc.

As previously noted, we made use of the corpus management and text analysis software called Sketch Engine. Thus, for the creation of our own corpus, we used the function enabled in this tool for this purpose, which can be found in the upper right corner of the dashboard on a button labeled "New corpus" after logging in. We then wrote the name of the corpus, indicated whether it is monolingual or multilingual, the language of the texts and provided a description. Once the characteristics were set, we added our 45 documents previously downloaded from Google Scholar and inserted them in PDF format with the "I have my own texts" button. When we had

added all the scientific publications, we compiled our corpus by clicking on the “Compile” button on the next screen in a simple and fast way.

3.3 Extraction of seed terms

Terminology extraction from our corpus has been performed with Sketch Engine through the “Wordlist. Frequency list” functionality (“Advanced” option). On the right-hand side, the settings are modified until the 50 most frequently used nouns in the 45 scientific publications are obtained. As soon as the characteristics have been adjusted, the “Go” button should be clicked to retrieve the 50 most frequent nouns and their absolute frequency, i.e., the number of times the element has been found in the corpus, as shown in Figure 1:

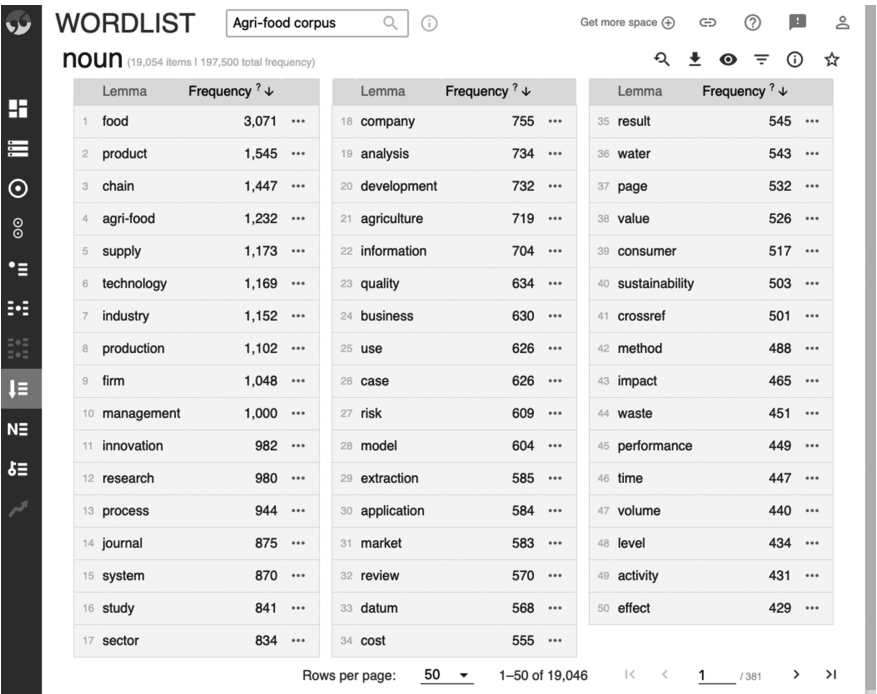


Figure 1: List of the 50 most frequent nouns extracted from our corpus on agri-foods.

Nouns, verbs, and adjectives are among the most important components of a discourse. Indeed, mastery of these units on a given topic will be essential for professional interpreters, as it will enable them to be more confident and at ease when

dealing with a specialized discourse. Hence, following the same procedure, we also retrieved the 50 verbs and 50 adjectives with the highest absolute frequency in our specialized corpus, which can be consulted in Figures 2 and 3 (see below).

These lists of the most frequent terminology (nouns, verbs, and adjectives) extracted from a specialized corpus are of particular interest to interpreters and translators not only from a linguistic standpoint, but also for the information that can be gleaned from them. For example, it is worth noting that words, such as *technology* or *innovation*, are found in the sixth and eleventh positions in the list of nouns. Likewise, it is also relevant that the verb *develop* is in the fifth position in the list of verbs, and adjectives such as *different* and *new* in the third and fourth position in the list of adjectives. This data is very valuable, since the professional or trainee interpreter can deduce from it that this is a sector that is innovating in its processes by means of new technologies.

The retrieval of multi-word expressions and sentences containing the most frequent terminology and these multi-word expressions is explained below by way of an example. The expressions and sentences concerned can be consulted through the gamified activities collected in this book chapter. We take as an illustration the word *food*, which is the most frequent noun. As can be appreciated in Figure 1 (see above), next to the absolute frequency, there are three dots at our disposal. Thanks to them, we can access the “Concordance” function and thus obtain sentences containing the noun *food*, which we then incorporate into our gamified activities. Figure 4 (see below) shows the list of some sentences containing the noun *food*.

In the same way, we use the “Word Sketch” option to choose multi-word expressions that we add later to our gamification exercises. Clicking again on the three dots, we can locate in sentences the multi-word expressions we need, which can be seen in Figure 5 (see below) for the case we are discussing.

As can be noticed in the lists compiled, it seems that the agri-food sector uses quite accessible terminology characteristic of a non-specialized language despite being a specialized field, which facilitates the task for interpreters and translators who want to work for this industry. Following these steps, we will recover the terminology, multi-word expressions and sentences that we consider most suitable for the practice and preparation of professional interpreters and trainees for a given assignment in the agri-food industry. Finally, multi-word expressions deserve special attention, since they would allow the practice of common collocations in the sector.

WORDLIST

Agri-food corpus

verb

11,851 items | 47,996 total frequency)

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Lemma	Frequency [?] ↓
1 be	10,119 ...
2 have	2,064 ...
3 use	1,334 ...
4 include	547 ...
5 develop	500 ...
6 increase	492 ...
7 base	484 ...
8 provide	471 ...
9 improve	396 ...
10 make	369 ...
11 reduce	359 ...
12 do	320 ...
13 show	312 ...
14 relate	298 ...
15 consider	282 ...
16 identify	282 ...
17 follow	274 ...

Lemma	Frequency [?] ↓
18 require	261 ...
19 find	253 ...
20 need	239 ...
21 cool	226 ...
22 apply	226 ...
23 produce	224 ...
24 lead	223 ...
25 affect	209 ...
26 take	206 ...
27 see	201 ...
28 focus	191 ...
29 obtain	189 ...
30 present	188 ...
31 allow	187 ...
32 create	186 ...
33 compare	185 ...
34 generate	175 ...

Lemma	Frequency [?] ↓
35 represent	174 ...
36 become	172 ...
37 support	171 ...
38 emerge	170 ...
39 give	169 ...
40 indicate	167 ...
41 regard	167 ...
42 involve	165 ...
43 dry	163 ...
44 associate	162 ...
45 achieve	158 ...
46 address	157 ...
47 implement	153 ...
48 operate	150 ...
49 process	148 ...
50 grow	146 ...

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Figure 2: List of the 50 most frequent verbs extracted from our corpus on agri-foods.

WORDLIST

Agri-food corpus

(3,985 items | 41,787 total frequency)

adjective

Lemma	Frequency [?] ↓
1 high	696 ...
2 environmental	661 ...
3 different	533 ...
4 new	514 ...
5 sustainable	514 ...
6 agricultural	509 ...
7 green	447 ...
8 low	358 ...
9 economic	342 ...
10 large	327 ...
11 important	324 ...
12 available	305 ...
13 rfid	303 ...
14 due	294 ...
15 social	290 ...
16 many	273 ...
17 public	272 ...

Lemma	Frequency [?] ↓
18 good	267 ...
19 specific	266 ...
20 small	254 ...
21 significant	247 ...
22 main	233 ...
23 online	233 ...
24 global	228 ...
25 natural	224 ...
26 future	215 ...
27 total	210 ...
28 industrial	196 ...
29 deep	196 ...
30 great	190 ...
31 several	185 ...
32 potential	184 ...
33 key	177 ...
34 same	174 ...

Lemma	Frequency [?] ↓
35 various	169 ...
36 local	168 ...
37 efficient	168 ...
38 recent	167 ...
39 international	165 ...
40 first	163 ...
41 possible	159 ...
42 relevant	157 ...
43 private	156 ...
44 eutectic	155 ...
45 human	155 ...
46 current	155 ...
47 organic	152 ...
48 critical	149 ...
49 major	142 ...
50 raw	140 ...

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Figure 3: List of the 50 most frequent adjectives extracted from our corpus on agri-foods.

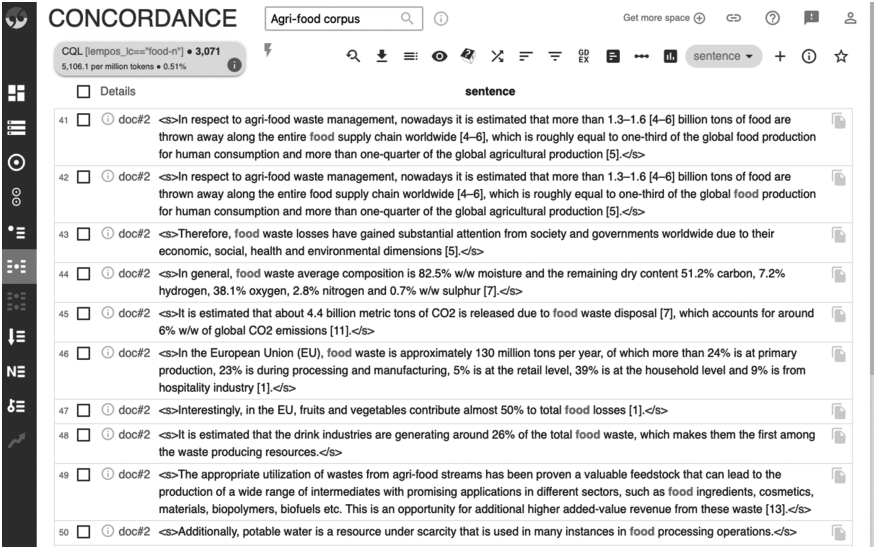


Figure 4: List of some sentences containing the noun “food”.

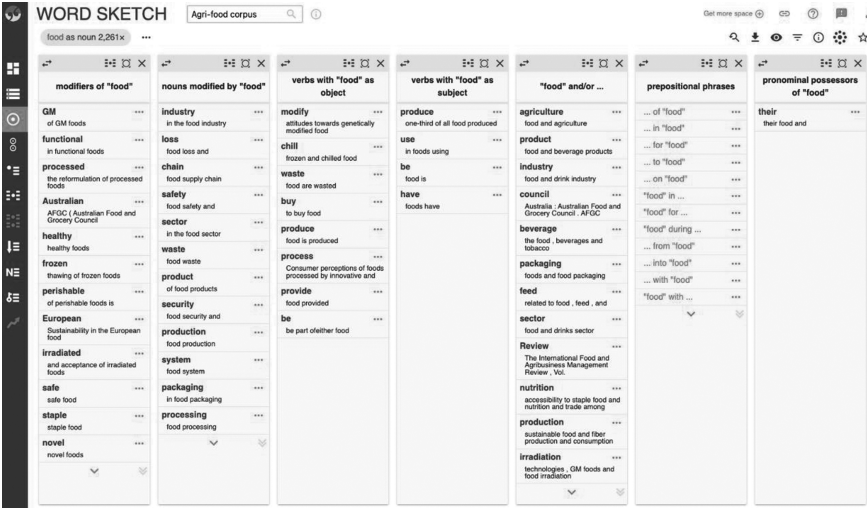


Figure 5: Word Sketch function for the noun “food”.

4 A structured set of terminology-orientated gamified activities

Below we present the set of seed-based gamified activities (GAs) that we have created according to the specifications and targets indicated in Section 3. For the design of these exercises, we have selected the mechanics, dynamics and components of gamification that we considered most appropriate among those offered by the chosen gamification platforms available on the web for the training of vocabulary about the agri-food industry and for the potential target audience. By means of these activities, it will be possible to work with this terminology contextualized in real texts compiled in the form of a corpus. These activities are available on the free and open access platforms where they have been developed and can be accessed through these platforms and through the links we share in this chapter.

On the one hand, we have chosen the Genially platform because it is an online tool that enables users to create and use interactive and visual content in a simple and fast way. This tool has a multitude of fully editable templates available and even a gamification templates section. Users can create their own design from scratch or modify existing templates as much as they want. This software contains very striking templates, sometimes inspired by existing games, TV shows, etc. that will arouse the curiosity and motivation of players. It is possible to introduce all the internal and external gamified dynamics, mechanics and components desired without any programming or design knowledge. This platform is especially interesting for the creation of activities for interpreters, as it allows the uploading of audios, YouTube videos, etc.

On the other hand, we have also selected Baamboozle, since it is a free tool to create games or to play available games designed by other users. The development of gamified activities is very simple. For each question you can enter words, sentences, images, memes, stickers or GIFs, which give the questions a more fun twist. With the free version, the gamified activity can be played with 1 to 4 teams, so it can also be played individually. In addition, visual elements can be added to the question numbers and power-ups, which can benefit or harm the different teams (win 5 points, lose 10 points, etc.), can be included. Moreover, the teacher can add more conditions orally before starting. This is a platform with many possibilities and is very motivating because of the internal and external elements that can be introduced in the different questions.

Regarding gamification platforms, Kahoot! is one of the most popular game-based platforms used not only in the educational environment, but also in the business world. Indeed, it has more than 70 million users actively using the platform per month (Wang & Tahir, 2020). However, more and more different gamifi-

cation platforms, such as Gimkit, Quizizz, Quizlet, Educaplay, Socrative, etc. are emerging and being applied in different sectors. The choice of one platform or another will depend on the objectives to be attained and the participants for whom the activity or gamified system is intended.

4.1 GA1: The lucky roulette wheel for interpreters

This gamified exercise has been designed from an existing template to which we have introduced modifications on the open access platform known as Genially. In this gamified activity, interpreters have to click on the roulette wheel to spin it and check how many points they will win if they choose the correct answer. When the roulette stops spinning, the interpreter can read a sentence in Spanish about agri-food. Our corpus is in English, so the Spanish sentence is a translation we propose from the original English sentence extracted from our corpus. This activity is created to practice especially some of the most frequent nouns retrieved from our corpus, but as we added complete sentences, there are also adjectives, verbs, and multi-word expressions that the interpreters will be able to work on.

This lucky roulette wheel consists in that the interpreters, after reading the sentence in Spanish, will have to listen to choices A and B in English as possible interpretation options, inserted in audio form using the voices of Google Translate and DeepL Translator, for the proposed sentence in Spanish. Only one of the options is correct and corresponds to an exact sentence taken from our corpus. In the incorrect option, we introduced grammatical errors, false meanings, etc.

Following the classification of Werbach and Hunter (2015), we used the next gamification elements: dynamics (emotions and progression, since users will be curious about a new exercise and will have to get the correct answer to move on to the next question); mechanics (challenges, opportunity, feedback, and rewards, because they will be challenged to get the maximum number of questions right and will have several opportunities to do so. Moreover, they will be checking their correct and wrong answers as they go along and at the end, they will pick a fictitious reward); and components (achievements, points, and virtual goods, as they will be able to select the correct interpretations and for each correct answer, they will win different virtual points. Finally, they can get a virtual prize). A sample of the activity can be seen in Figure 6.

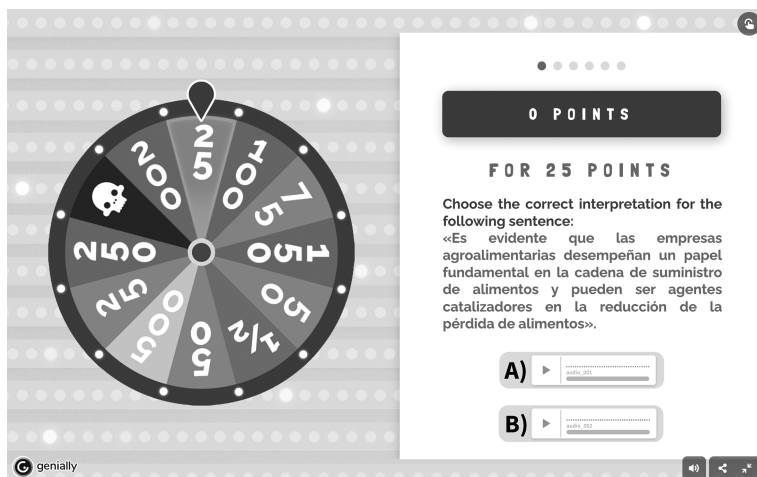


Figure 6: First question of GA1: The lucky roulette wheel for interpreters.²

4.2 GA2: Think fast and interpret!

This gamification exercise has been created to work mainly on some of the most frequently used adjectives in the agri-food field. Nevertheless, by incorporating real complete sentences obtained from our corpus, the interpreter will also be able to work on other interesting terminology. In this gamified activity, the user acquires an active role that will be very useful when they have to cope with a possible interpreting assignment.

It has been developed on Baamboozle, a web application for creating games for free. This activity can be played alone or in teams. Users will find a virtual board with squares numbered from 1 to 14. To start practicing, the interpreters will have to randomly select squares. By clicking on any box, an English sentence related to the agri-food sector will appear. At that moment, the interpreter or the team playing the game will have to interpret the phrase aloud in Spanish and click on the “Check” button below to check the answer. If they have given the same or a similar answer, they must click on “Okay!” to get the points. On the contrary, if they make a mistake, they will have to choose the “Oops!” option and will not score any points. Then, it will be the other team’s turn or, in the case of a single player, they will have to choose another square.

² <https://view.genial.ly/638dde21083ef100195f28c7/interactive-content-the-lucky-roulette-wheel-for-interpretersmarta-alcaide-martinez> (Retrieved May 10, 2023).

In Think fast and interpret! we have implemented the following gamified elements: dynamics (emotions, progression, and relationships between classmates, since it is a striking game in which progress is made in turns and teamwork is essential); mechanics (challenges, opportunity, competition, cooperation, feedback, rewards, and turns, as the interpreter is put to the test and has a single opportunity for each square. In the case of the team game, a competition is created in which the teams have to work coordinated in turns to win, while at the same time checking their hits and misses on each question. A real or fictitious reward could be awarded to the winning team); and components (achievements, points, and teams, because the interpreters get to prepare for a specialization topic and for each correct interpretation, they add points to their team's scoreboard).

This website also enables the activation of power-ups (to steal points from the other team, gain points without answering any questions, etc.), which can be very motivating and arouse the competitiveness of the participants. We have also included related GIFs in each question to give it a more attractive and playful appearance. A sample of the exercise can be found in Figure 7.

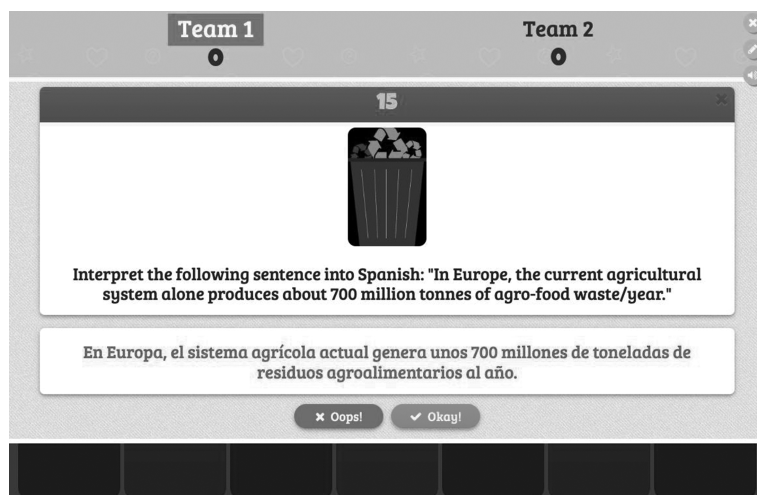


Figure 7: Random square of GA2: Think fast and interpret!.³

³ <https://www.baamboozle.com/classic/1320304> (Retrieved May 10, 2023).

4.3 GA3: Interbomb

This gamified exercise has been prepared by modifying an existing template on the Genially platform. As can be appreciated in the aesthetics of the game, Interbomb is inspired by the Spanish TV quiz show known as ¡Boom! which has a similar dynamic to this exercise. This activity has been conceived to encourage the practice of the most frequent verbs of our corpus used in the agri-food industry. Interpreters will find two types of questions:

- (1) the interpreter(s) will be able to read a sentence in which there is a gap that they must fill in with the appropriate verb. They will have to choose between four possible options, which are the four wires of the bomb and are in audio form);
- (2) the interpreter(s) will have to listen to an audio available at the top containing a sentence in English about agri-food. Then, they will have to select the correct interpretation into Spanish, after listening to the four available audio options. They will have to be very attentive, as only small details change between the options.

It should be noted that, as in the other activities, by working with full sentences taken from our corpus, interpreters can also learn other domain-specific terminology and other skills. As our corpus is in English, the interpretations into Spanish have been elaborated by us, although we have used the voices of Google Translate and DeepL Translator for the recordings.

The gamification elements we have employed are: dynamics (emotions and progression, since the players will be more motivated and concentrated than with traditional exercises and will have to pick the correct answer to continue advancing); mechanics (challenges, opportunity, competition, cooperation, feedback, rewards, and turns, as the participants will have to be very focused to get the right answer and will have several opportunities to achieve this. Interbomb can be played by a single person or by several teams. If played in teams, a competition will be set up in which the teams will have to be united and respect the turns to get the final agreed reward. Users will receive immediate feedback for each question); and components (achievements and teams, because individually or in teams they will have to select the correct options to become the winners of the activity). Figures 8 and 9 show a sample of the exercise.

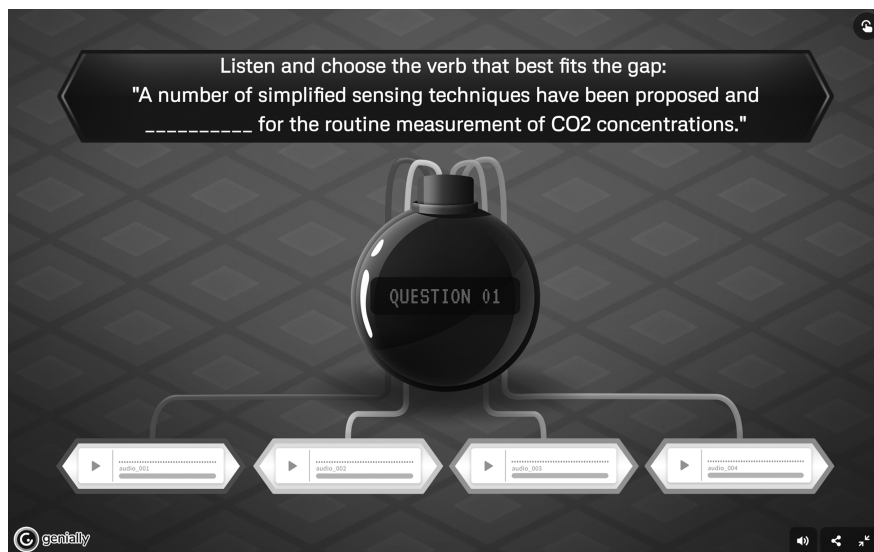


Figure 8: Question 1 Type 1 of GA3: Interbomb.⁴

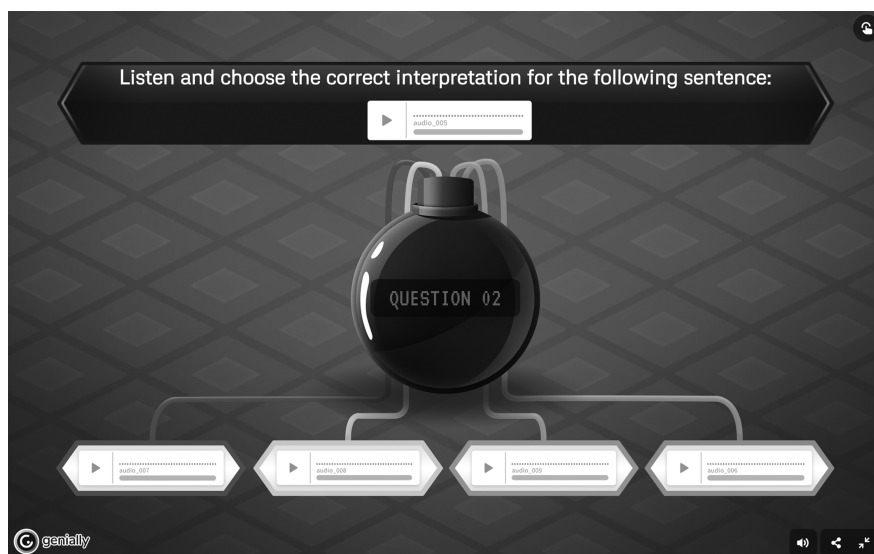


Figure 9: Question 2 Type 2 of GA3: Interbomb.

⁴ <https://view.genial.ly/638f1e7f0f5aa900118b5519/interactive-content-interbombmarta-alcaide-martinez> (Retrieved May 10, 2023).

5 Conclusions

Today, interpreters have hardly any innovative tools to carry out the preparation phase before an assignment. In this chapter, we offer three gamified activities for professional interpreters, trainees and interpreting instructors that are available on the open access platforms Genially and Baamboozle. At this point we would like to mention that, on the Baamboozle platform, users can see how many times their activities are played. Our Think fast and interpret! activity has been published for only a month and has already been played 19 times, which means that gamification is already starting to have an impact in the area of interpretation.

This result encourages us to continue creating gamified exercises, which we will be able to design by further exploiting the corpus we have compiled. Likewise, as future work, we would like to conduct an empirical study with interpreting students with the designed gamified exercises to test the benefits of using gamification in the preparation phase of the interpreters as opposed to the traditional preparation by means of glossaries and bilingual lists. It would be interesting to divide a group of students into two subgroups and compare the degree of quality achieved in the interpretations provided by the group that has prepared with the gamified activities and the group that has prepared with traditional methods.

Throughout this paper, we have shed light on a relevant economic, environmental and social sector and we have emphasized the benefits of linking this industry with interpreting and gamification. We have also made accessible to the academic world a series of gamified and technological activities that will help interpreters to train for a specialized assignment on agri-food. In addition, the production of these activities has been made feasible thanks to the domain-specific ad hoc corpus of 45 scientific articles on agri-foods that we have previously compiled in Sketch Engine and from which we have retrieved real, non-adapted sentences.

The memorization of domain-specific terminology is one of the major challenges that interpreters have to face in their daily work. Thus, this chapter presents a novel method of preparation that aims to facilitate the difficult work that interpreters have to do during the preparation phase. As far as we know, this is the first research study to propose a gamification-based methodology for the preparation phase that both students and professionals of interpreting in its different modalities have to carry out. In this way, we hope that this chapter will serve as an inspiration to other researchers, lecturers and interpreters.

Finally, even though our gamified activities have been designed for interpreters, they could also be beneficial for professional translators and translation students. These exercises focus mainly on terminology practice in the specialized subject area of agri-food. Therefore, they could help translators do their work faster, without having to spend a lot of time on the task of documentation or

searching for equivalents. Ultimately, these activities could also be used by interpreting and translation trainers to get their students to work on domain-specific terminology through playful activities, as well as by academics researching how to innovate in this field.

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